

HAZEL KOO

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SUMMARY

Machine Learning Engineer specializing in LLMs, RAG, and knowledge-grounded AI, with hands-on experience building and evaluating end-to-end ML systems. Experienced in graph-based retrieval, knowledge graphs, LLM fine-tuning, and inference optimization, with publications including AAAI 2024.

PUBLICATIONS

- Duplicate Elimination on Active Memory for Self-Supervised Class-Imbalanced Learning, **AAAI 2024**
- OmEGa(Q): Ontology-based Information Extraction Framework for Constructing Task-Centric Knowledge Graph from Manufacturing Documents with Large Language Model, **Advanced Engineering Informatics 2025**

EDUCATION

University of Texas at Dallas

Master of Data Science and Statistics / Concentration: Applied Statistics

Aug. 2025 – May. 2027

- NSM McDermott Master's Student Fellowship (2025-2027)
- **Winner, Best Use of Solana** - WeHack 2026 (MLH-affiliated hackathon, 300+ participants)

Yonsei University

Master of Artificial Intelligence (GPA: 4.15 / 4.5)

Sept. 2023 – Aug. 2025

- Teaching Assistant, "Computer Programming" (Fall 2024); Admission Scholarship (Fall 2023)

Kyung Hee University

Bachelor of Computer Science and Engineering (GPA: 3.91 / 4.5)

Mar. 2018 – Feb. 2023

- Academic Excellence Scholarship (Spring 2018); Software Scholarship (Spring 2022)

SKILLS

Languages: Python, SQL, R, C++, JavaScript (React, Node.js), Java, MATLAB, HTML/CSS

ML/AI: PyTorch, Hugging Face Transformers, scikit-learn, LLMs, RAG, Knowledge Graphs, CUDA

Tools & Platforms: AWS (EC2, S3), Claude Code, Linux, Git, OpenAI API, Protégé, GraphDB, Cursor

EXPERIENCE

LG AI RESEARCH

Research Scientist Intern | Data Intelligence Lab

May. 2026 – Aug. 2026

- Designed and implemented **BindRAG**, a graph-based RAG framework using programmatic query decomposition and explicit/implicit entity binding, outperforming leading baselines by **+6.5 / +19.0 / +5.5 pp** in string accuracy on 3 benchmarks, respectively; first-author manuscript in preparation for ICLR 2027.
- Reproduced and benchmarked **9 graph-based RAG methods across 3 multi-hop QA datasets** using the DIGIMON evaluation framework, analyzing retrieval and reasoning failures to identify limitations in existing approaches and guide the development of BindRAG.
- Tuned LLM reasoning configurations in an internal RAG platform, reducing inference latency by **~40% with no degradation** in answer accuracy; identified and categorized mislabeled samples through evaluation data analysis.

Yonsei University

Research Assistant | Internet Computing Laboratory

Aug. 2023 – Aug. 2025

- Built an **ontology-based information extraction pipeline** integrating LLMs with a domain-specific knowledge graph for Hyundai Motor Company, constructing a KG with **7 entity classes and 22 relation types from 422 automotive documents** and achieving **93.7% coverage across 684 competency questions**.
- Fine-tuned **LLaMA-based LLMs** using the constructed KG and evaluated 5 event argument extraction baselines, achieving a **+66% F1 score improvement** over state-of-the-art models.
- Developed a **RAG-based** complex claim fact-checking framework leveraging advanced **prompt engineering** using GPT-3.5-turbo, outperforming NLI fine-tuned baselines by **+7.92% F1 score** and contributing to a **research** an **ongoing patent** filing.

Seoul National University

Research Intern | Bio-Intelligence Laboratory

Jan. 2023 – Apr. 2023

- Developed and evaluated representation-learning methods for class-imbalanced learning, contributing to a co-authored paper published at **AAAI 2024**.